

CS5275 – The Algorithm Designer's Toolkit
(S2 AY2025/26)

Lecture 17:

Multiplicative weights update – applications

Learning a linear classifier

- Given m labeled examples: $(\mathbf{a}_1, \ell_1), (\mathbf{a}_2, \ell_2), \dots, (\mathbf{a}_m, \ell_m)$
 - $\mathbf{a}_j \in \mathbb{R}^n$ is a feature vector
 - $\ell_j \in \{-1, 1\}$ is a label.

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- Suppose there exists a vector $\mathbf{x} \in \mathbb{R}^n$ with $x_i \geq 0$ such that $\text{sgn}(\mathbf{a}_j \cdot \mathbf{x}) = \ell_j$.
- **Goal:** Find a vector $\mathbf{x} \in \mathbb{R}^n$ with $x_i \geq 0$ such that $\text{sgn}(\mathbf{a}_j \cdot \mathbf{x}) = \ell_j$.

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Without loss of generality,
we may assume:

- $\mathbf{x} \cdot \mathbf{1} = 1$
- $\ell_j = 1$

$\mathbf{1}$ is the all-1 vector.

If this is not the case, then we may replace $\mathbf{a}_j$ with $-\mathbf{a}_j$.

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LP formulation:

- $\mathbf{a}_j \cdot \mathbf{x} > 0 \quad \forall j \in [m]$
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Learning a linear classifier

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Claim: Suppose a large-margin solution $\mathbf{x}^*$ exists:

$$\mathbf{a}_j \cdot \mathbf{x}^* \geq \epsilon \quad \forall j \in [m]$$

Then there is an efficient algorithm that finds a feasible solution $\mathbf{x}$.

Connection to MWU

n coordinates

Multiplicative weights update:

- Assign initial weights to the experts to be identical weights.
- Update these weights multiplicatively and iteratively according to how well the experts performed.
- In each iteration, we decide the expert to follow based on the weighted distribution.

A candidate solution $\mathbf{x} \in \mathbb{R}^n$

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If all constraints $\mathbf{a}_j \cdot \mathbf{x} > 0$ are satisfied, then the problem is solved.

Otherwise, select an unsatisfied constraint and assign a cost to each coordinate based on how severely this constraint is violated.

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MWU performs competitively against any convex combination of experts.

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Algorithm

Multiplicative weights algorithm:

- Fix $0 < \eta \leq \frac{1}{2}$.
- Assign initial weights to the experts to be identical weights:
 - $w_i^{(1)} = 1$ for each expert $i \in \{1, \dots, n\}$.
- In each iteration $1 \leq t \leq T$:
 - Let $\Phi^{(t)} = \sum_{i=1}^n w_i^{(t)}$.
 - Select a distribution $\mathbf{p}^{(t)} = \left(\frac{w_1^{(t)}}{\Phi^{(t)}}, \frac{w_2^{(t)}}{\Phi^{(t)}}, \dots, \frac{w_n^{(t)}}{\Phi^{(t)}} \right)$.
 - Update the weight for every expert: $w_i^{(t+1)} = \left(1 + \frac{\eta m_i^{(t)}}{\rho} \right) w_i^{(t)}$.

This is seen as candidate solution $\mathbf{x} \in \mathbb{R}^n$.

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- If $\mathbf{x} = \mathbf{p}^{(t)}$ is already feasible, then we are done.
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Run the multiplicative weights algorithm in the **gain** form with $m_i^{(t)} \in [-\rho, \rho]$.

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Suppose we are always in this case:

Analysis

$$0 \geq \sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)}$$

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$$\text{Recall: } \sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \geq \sum_{t=1}^T (\mathbf{m}^{(t)} - \eta |\mathbf{m}^{(t)}|) \cdot \mathbf{p}^* - \frac{\rho \ln n}{\eta}$$

the expected gain of the algorithm

$$\mathbf{m}^{(t)} = (m_1^{(t)}, m_2^{(t)}, \dots, m_n^{(t)})$$

- $m_i^{(t)} \in [-1, 1]$ is the cost for expert i in iteration t .

$\mathbf{p}^{(t)}$ = distribution of experts in iteration t .

$\mathbf{p}^*$ = any convex combination of experts.

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 &\geq T(\epsilon - \rho\eta) - \frac{\rho \ln n}{\eta}
 \end{aligned}$$

$$m_i^{(t)} \in [-\rho, \rho] \rightarrow |\mathbf{m}^{(t)}| \cdot \mathbf{x}^* \leq \rho$$

$$\mathbf{a}_j \cdot \mathbf{x}^* \geq \epsilon \rightarrow \mathbf{m}^{(t)} \cdot \mathbf{x}^* \geq \epsilon$$

Recall: $\sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \geq \sum_{t=1}^T (\mathbf{m}^{(t)} - \eta |\mathbf{m}^{(t)}|) \cdot \mathbf{p}^* - \frac{\rho \ln n}{\eta}$

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Impossible

Suppose we are always in this case:

Analysis

$$\begin{aligned} 0 &\geq \sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \\ &\geq \sum_{t=1}^T (\mathbf{m}^{(t)} - \eta |\mathbf{m}^{(t)}|) \cdot \mathbf{x}^* - \frac{\rho \ln n}{\eta} \\ &\geq T(\epsilon - \rho\eta) - \frac{\rho \ln n}{\eta} > 0 \end{aligned}$$

Selecting $\eta = \frac{\epsilon}{2\rho}$ and $T = 1 + \left\lceil \frac{4\rho^2 \ln n}{\epsilon^2} \right\rceil$

Run MWU with these parameters.



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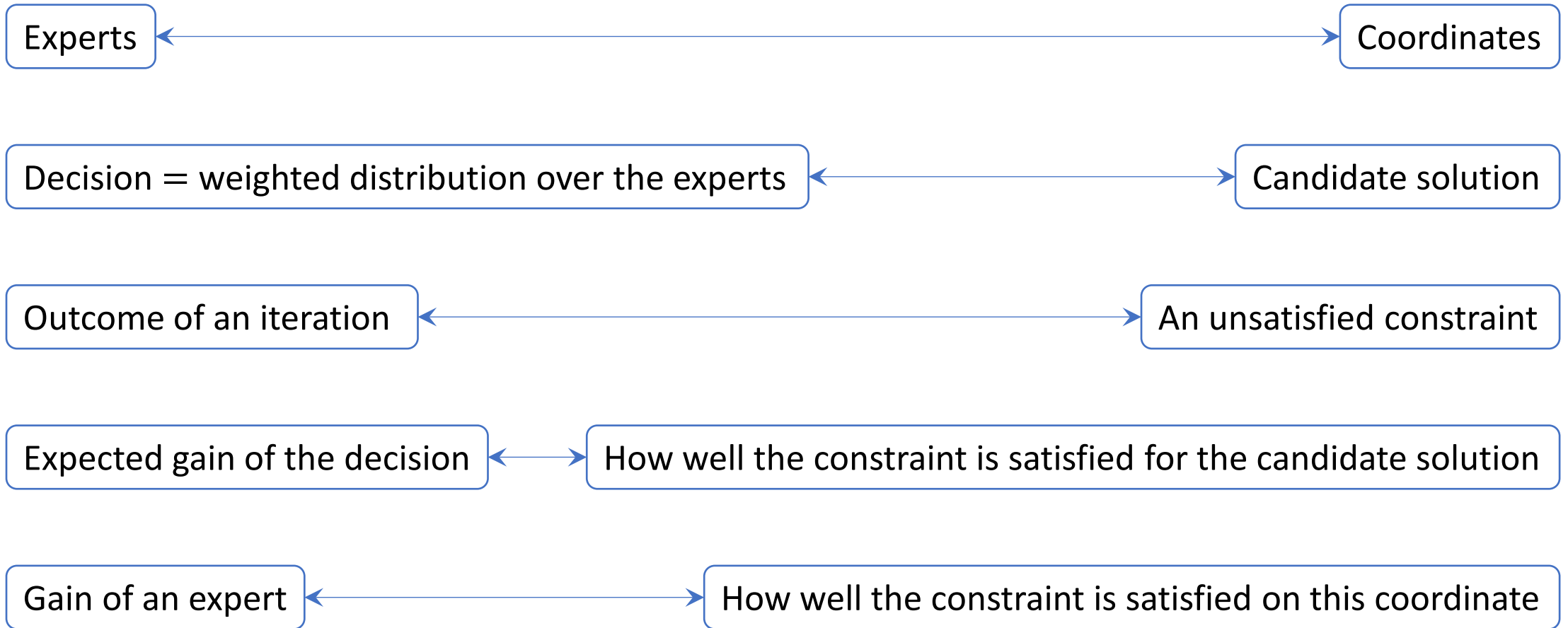
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Summary of the main ideas



Zero-sum games

- Let $\mathbf{A}$ be the payoff matrix of a finite 2-player zero-sum game.

$\mathbf{A}$ is an $(n \times n)$ matrix.

$A(i, j) \in [-1, 1]$ specifies the payoff when the row player plays strategy $i \in [n]$ and the column player plays strategy $j \in [n]$.

If we do not have $A(i, j) \in [-1, 1]$ for all i and j , we may rescale them.

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	Rock	Paper	Scissors
Rock	0	1	-1
Paper	-1	0	1
Scissors	1	-1	0

Example: rock-paper-scissors

Zero-sum games

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

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$A(i, j) \in [-1, 1]$ specifies the payoff when the row player plays strategy $i \in [n]$ and the column player plays strategy $j \in [n]$.

Suppose the row player chooses the **mixed strategy**

$$\mathbf{p} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2} \right).$$

A probability distribution over the strategies

A single strategy

The best response **pure strategy** of the column player is

$j = \text{rock}.$

The expected payoff is

$$A(\mathbf{p}, j) = \mathbf{E}_{i \sim \mathbf{p}}[A(i, j)] = \frac{1}{4} \cdot 0 + \frac{1}{4} \cdot (-1) + \frac{1}{2} \cdot 1 = \frac{1}{4}.$$

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Example 2

Goals:

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- Row player wants to minimize the payoff.

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$A(i, j) \in [-1, 1]$ specifies the payoff when the row player plays strategy $i \in [n]$ and the column player plays strategy $j \in [n]$.

The best **mixed strategy** for the row player is

$$\mathbf{p} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right).$$

A probability distribution over the strategies

A single strategy

Regardless of the column **pure strategy** j , we have

$$A(\mathbf{p}, j) = \mathbf{E}_{i \sim \mathbf{p}}[A(i, j)] = 0.$$

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Example: rock-paper-scissors

Max–min inequality

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

$$\min_p \max_q A(\mathbf{p}, \mathbf{q}) = \min_p \max_j A(\mathbf{p}, j) \geq \max_q \min_i A(i, \mathbf{q}) = \max_q \min_p A(\mathbf{p}, \mathbf{q})$$

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Responding with a
mixed strategy $\mathbf{q}$.

Responding with a
pure strategy j .

Similar

Intuition: For the player that decides its strategy later, **randomness** does not help.

The row player first announces its mixed strategy $\mathbf{p}$.
The column player decides the **best response** to $\mathbf{p}$.

Max–min inequality

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$$\max_j A(\mathbf{p}, j) \geq \max_{\mathbf{q}} A(\mathbf{p}, \mathbf{q})$$

$$\max_j A(\mathbf{p}, j) \leq \max_{\mathbf{q}} A(\mathbf{p}, \mathbf{q})$$

Each pure strategy j is
also a mixed strategy $\mathbf{q}$.

For each mixed strategy $\mathbf{q}$, there exists a pure strategy j such that $A(\mathbf{p}, j) \geq A(\mathbf{p}, \mathbf{q})$.

$A(\mathbf{p}, \mathbf{q})$ is a convex combination of $A(\mathbf{p}, j)$ over all strategies j .

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Intuition: The player that decides its strategy **later** has advantage.

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The row player decides the best response i to $\mathbf{q}$.

Intuition: The player that decides its strategy **later** has advantage.

Max–min inequality

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p}, \mathbf{q}) = \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) \geq \max_{\mathbf{q}} \min_i A(i, \mathbf{q}) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p}, \mathbf{q})$$

Select $\mathbf{p}^*$ and $\mathbf{q}^*$ to be the best row and column mixed strategies:

- $\max_{\mathbf{q}} A(\mathbf{p}^*, \mathbf{q}) = \min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p}, \mathbf{q})$
- $\min_{\mathbf{p}} A(\mathbf{p}, \mathbf{q}^*) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p}, \mathbf{q})$

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p}, \mathbf{q}) = \max_{\mathbf{q}} A(\mathbf{p}^*, \mathbf{q}) \geq A(\mathbf{p}^*, \mathbf{q}^*) \geq \min_{\mathbf{p}} A(\mathbf{p}, \mathbf{q}^*) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p}, \mathbf{q})$$

Minimax theorem

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Minimax theorem:

$$\lambda^* = \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \max_{\mathbf{q}} \min_i A(i, \mathbf{q})$$

Definition of λ^*

Minimax theorem

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Definition of λ^*

Claim: There is an efficient algorithm that estimates λ^* within an additive $\pm \epsilon$ error and finds near-optimal mixed strategies $\tilde{\mathbf{p}}$ and $\tilde{\mathbf{q}}$ of the two players.

- $\lambda^* - \epsilon \leq \min_i A(i, \tilde{\mathbf{q}})$
- $\max_j A(\tilde{\mathbf{p}}, j) \leq \lambda^* + \epsilon.$

Minimax theorem

Goals:

- Column player wants to **maximize** the payoff.
- Row player wants to **minimize** the payoff.

$$\min_{\mathbf{p}} \max_{\mathbf{q}} A(\mathbf{p}, \mathbf{q}) = \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) \geq \max_{\mathbf{q}} \min_i A(i, \mathbf{q}) = \max_{\mathbf{q}} \min_{\mathbf{p}} A(\mathbf{p}, \mathbf{q})$$

Minimax theorem:

$$\lambda^* = \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \max_{\mathbf{q}} \min_i A(i, \mathbf{q})$$

Definition of λ^*

$\epsilon \rightarrow 0$

$$\begin{aligned} & \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) - \epsilon \\ &= \lambda^* - \epsilon \\ &\leq \min_i A(i, \tilde{\mathbf{q}}) \\ &\leq \max_{\mathbf{q}} \min_i A(i, \mathbf{q}) \end{aligned}$$

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Algorithm

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

Multiplicative weights algorithm:

- Fix $0 < \eta \leq \frac{1}{2}$.
- Assign initial weights to the experts to be identical weights:
 - $w_i^{(1)} = 1$ for each expert $i \in \{1, \dots, n\}$.
- In each iteration $1 \leq t \leq T$:
 - Let $\Phi^{(t)} = \sum_{i=1}^n w_i^{(t)}$.
 - Select a distribution $\mathbf{p}^{(t)} = \left(\frac{w_1^{(t)}}{\Phi^{(t)}}, \frac{w_2^{(t)}}{\Phi^{(t)}}, \dots, \frac{w_n^{(t)}}{\Phi^{(t)}} \right)$.
 - Update the weight for every expert: $w_i^{(t+1)} = (1 - \eta m_i^{(t)}) w_i^{(t)}$.

Treat each row pure strategy $\{1, 2, \dots, n\}$ as an expert.

This is seen as a row mixed strategy.

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Algorithm

Goals:

- Column player wants to maximize the payoff.
- Row player wants to minimize the payoff.

Intuitively:

MWU performs competitively against any convex combination of experts.

→ If we run the algorithm for many iterations, then on average, the payoff of a row mixed strategy $\mathbf{p}^{(t)}$ is very close to λ^* .

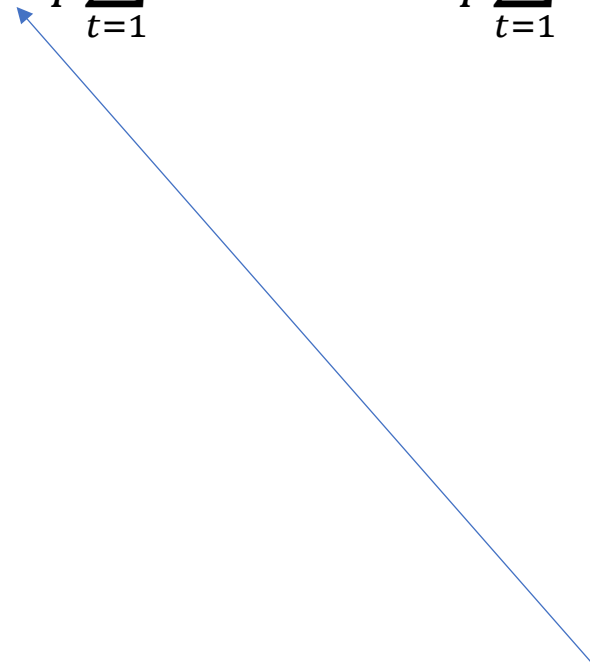
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Analysis

$$\lambda^* \leq \frac{1}{T} \sum_{t=1}^T A(\mathbf{p}^{(t)}, j^{(t)}) = \frac{1}{T} \sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)}$$


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$\mathbf{p}'$ is any **row** mixed strategy

Analysis

$$\lambda^* \leq \frac{1}{T} \sum_{t=1}^T A(\mathbf{p}^{(t)}, j^{(t)}) = \frac{1}{T} \sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \leq \frac{\ln n}{\eta T} + \eta + \frac{1}{T} \sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}'$$

Recall: $\sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \leq \underbrace{\sum_{t=1}^T (\mathbf{m}^{(t)} + \eta |\mathbf{m}^{(t)}|)}_{\leq \eta T + \sum_{t=1}^T \mathbf{m}^{(t)} \cdot \mathbf{p}^*} \cdot \mathbf{p}^* + \frac{\rho \ln n}{\eta}$

$$\rho = 1$$

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$$= \frac{\ln n}{\eta T} + \eta + \frac{1}{T} \sum_{t=1}^T A(\mathbf{p}', j^{(t)})$$

$$\eta = \frac{\epsilon}{2}$$

$$T = \lceil 4\epsilon^{-2} \ln n \rceil$$



$$\leq \epsilon + \frac{1}{T} \sum_{t=1}^T A(\mathbf{p}', j^{(t)})$$

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$$\leq \epsilon + \frac{1}{T} \sum_{t=1}^T A(\mathbf{p}', j^{(t)})$$

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$$\mathbf{p}' = \mathbf{p}^*$$

If optimal **row** mixed strategy $\mathbf{p}^*$ is used, then $\mathbf{m}^{(t)} \cdot \mathbf{p}^* = A(\mathbf{p}^*, j^{(t)}) \leq \max_j A(\mathbf{p}^*, j) = \min_{\mathbf{p}} \max_j A(\mathbf{p}, j) = \lambda^*$

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$$\eta = \frac{\epsilon}{2}$$

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Near-optimal **row mixed strategy:** There exists $t \in [T]$ such that $\max_j A(\mathbf{p}^{(t)}, j) = A(\mathbf{p}^{(t)}, j^{(t)}) \leq \lambda^* + \epsilon$.

Row player wants to **minimize** the payoff.

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$$\leq \epsilon + \frac{1}{T} \sum_{t=1}^T A(\mathbf{p}', j^{(t)})$$

$$= \epsilon + A(\mathbf{p}', \tilde{\mathbf{q}})$$



$\tilde{\mathbf{q}}$ is defined as the distribution that assigns j with probability $\frac{|\{t : j^{(t)} = j\}|}{T}$.

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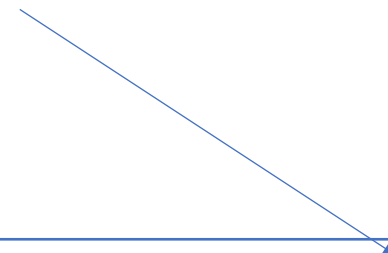


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Near-optimal **column** mixed strategy: $\lambda^* - \epsilon \leq A(\mathbf{p}', \tilde{\mathbf{q}}) = \min_i A(i, \tilde{\mathbf{q}})$



The analysis works for any $\mathbf{p}'$.

References

- Section 3 of “The Multiplicative Weights Update Method: a Meta-Algorithm and Applications” by Sanjeev Arora, Elad Hazan, and Satyen Kale
 - <https://theoryofcomputing.org/articles/v008a006/>